

Logic Building Assignment : 34

All below questions are depends on ASCII values of characters. Please consider below table to solve the questions.

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	00	Null	32	20	Space	64	40	@	96	60	`
1	01	Start of heading	33	21	!	65	41	A	97	61	a
2	02	Start of text	34	22	"	66	42	B	98	62	b
3	03	End of text	35	23	#	67	43	C	99	63	c
4	04	End of transmit	36	24	\$	68	44	D	100	64	d
5	05	Enquiry	37	25	%	69	45	E	101	65	e
6	06	Acknowledge	38	26	&	70	46	F	102	66	f
7	07	Audible bell	39	27	'	71	47	G	103	67	g
8	08	Backspace	40	28	(	72	48	H	104	68	h
9	09	Horizontal tab	41	29	)	73	49	I	105	69	i
10	0A	Line feed	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage return	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	47	2F	/	79	4F	O	111	6F	o
16	10	Data link escape	48	30	0	80	50	P	112	70	p
17	11	Device control 1	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	50	32	2	82	52	R	114	72	r
19	13	Device control 3	51	33	3	83	53	S	115	73	s
20	14	Device control 4	52	34	4	84	54	T	116	74	t
21	15	Neg. acknowledge	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	54	36	6	86	56	V	118	76	v
23	17	End trans. block	55	37	7	87	57	W	119	77	w
24	18	Cancel	56	38	8	88	58	X	120	78	x
25	19	End of medium	57	39	9	89	59	Y	121	79	y
26	1A	Substitution	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	60	3C	<	92	5C	\	124	7C	
29	1D	Group separator	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	63	3F	?	95	5F	_	127	7F	□

1. Write a program which displays ASCII table. Table contains symbol, Decimal, Hexadecimal and Octal representation of every member from 0 to 255.

```
void DisplayASCII()
{
    // Logic
}

int main()
{
    DisplayASCII();

    return 0;
}
```

2. Accept character from user. If character is small display its corresponding capital character, and if it small then display its corresponding capital. In other cases display as it is.

Input : Q

Output : q

Input : m

Output : M

Input : 4

Output : 4

Input : %

Output : %

```
void Display(char ch)
{
    // Logic
}
```

```
int main()
{
    char cValue = '\0';
}
```

```
printf("Enter the character");
scanf("%c",&cValue);
```

```
Display(cValue);
```

```
return 0;
```

```
}
```

3. Accept character from user. If it is capital then display all the characters from the input characters till Z. If input character is small then print all the characters in reverse order till a. In other cases return directly.

Input : Q

Output : Q R S T U V W X Y Z

Input : m

Output : m l k j i h g f e d c b a

Input : 8

Output :

```
void Display(char ch)
{
```

```
}
```

```
int main()
```

```
{
```

```
    char cValue = '\0';
```

```
    printf("Enter the character");
    scanf("%c",&cValue);
```

```
    Display(cValue);
```

```
    return 0;
```

```
}
```

4. Accept Character from user and check whether it is special symbol or not (!, @, #, \$, %, ^, &, *).

Input : %

Output : TRUE

Input : d

Output : FALSE

```
BOOL ChkSpecial(char ch)
{
    // Apply condition to check whether it is special or not.
}

int main()
{
    char cValue = '\0';
    BOOL bRet = FALSE;

    printf("Enter the character");
    scanf("%c",&cValue);

    bRet = ChkSpecial(cValue);

    if(bRet == TRUE)
    {
        printf("It is special Character");
    }
    else
    {
        printf("It is not a special Character");
    }

    return 0;
}
```

5. Accept character from user and display its ASCII value in decimal, octal and hexadecimal format.

Input : A

Output : Decimal 65

Octal
Hexadecimal

0101
0X41

```
void Display (char ch)
{
    // Logic
}

{
    char cValue = '\0';

    printf("Enter the character");
    scanf("%c",&cValue);

    Display(cValue);

    return 0;
}
```

